

Product Requirements Document

Bhasha Setu — Offline AI Vernacular Classroom Copilot for Mother-Tongue-Based Primary Education

SIH Problem Statement: 26042
Organisation: Government of Jharkhand, Department of Higher & Technical Education
Category / Theme: Software / Smart Education
Document version: 1.0 (pre-prototype baseline)
Status: Implementation-ready specification. All performance figures in this document are *engineering targets*, not measured results.

How to read this document

Every non-obvious technical claim is tagged:

Tag	Meaning
[VERIFIED]	Confirmed against a public, linked primary source (model card, repo, government document).
[ASSUMPTION]	An engineering assumption the team is choosing to make. Must be tested.
[VALIDATE]	Requires validation during prototyping. Do not present as fact in the SIH pitch.
[RESEARCH]	Open question. No public answer found. Needs original work or expert input.

No benchmark number, accuracy percentage, latency measurement, NIPUN code, dataset, or partnership in this document is invented. Where a number appears as a target, it is labelled as a target and paired with a measurement method.

1. Executive Summary

Jharkhand's PALASH programme places five tribal languages (Kurukh, Santhali, Mundari, Ho, Kharia) into primary classrooms, but the teachers posted in those classrooms are overwhelmingly Hindi-medium trained. The bottleneck is not curriculum and not policy — it is that a specific teacher, standing in front of a specific Grade 1 class on a specific morning, cannot say "open your books to the picture of the goat" in a language the children understand.

Bhasha Setu is a teaching instrument, not a translator. It is an offline Android application that lets a Hindi-medium teacher deliver an FLN lesson in a child's mother tongue: it hands the teacher a verified bilingual lesson script with native-speaker audio, translates live classroom speech in both directions within the 3-second constraint, and generates bilingual worksheets and flashcards

traceable to a NIPUN Bharat learning outcome. Everything runs with the tablet in airplane mode after one sync.

The three decisions that define this product:

1. **Santali (OI Chiki) is the MVP language.** It is the only one of the three named languages with published open models across all three required capabilities — machine translation, ASR, and TTS — because it is a Scheduled Language and therefore inside the AI4Bharat/Bhashini coverage envelope. Ho and Mundari have effectively no published open speech models. Section 15.1 gives the full evidence table. Pretending otherwise would produce a demo that fails on stage.
2. **Correctness is architectural, not statistical.** A neural model that confidently produces a wrong Santali word in a Grade 1 literacy lesson is worse than no product, because the teacher cannot detect the error and the child learns it. Every tribal-language string the app can *speak or print* is therefore drawn from a **verified content store** — human-reviewed, versioned, state-published — with the neural path used only for unverified live speech, and clearly marked as unverified when used.
3. **The 3-second requirement is met by not running a full neural pipeline most of the time.** Classroom speech is highly repetitive. A phrase bank of pre-translated, pre-synthesised classroom utterances handles the common case with cached audio playback (fast path); the ASR→MT→TTS chain handles the tail (dynamic path). The fast path is what gets demoed; the dynamic path is what makes it honest.

What is genuinely hard, stated up front: there is no publicly available on-device Santali TTS voice. Indic Parler-TTS supports Santali but is far too large for a 2 GB tablet. Closing that gap — a small VITS-class Santali voice, or a fully cached audio strategy — is the single largest technical risk in this project and is treated as such throughout (Sections 16.4, 38).

2. Problem Definition

2.1 The primary problem

A Hindi-medium-trained teacher assigned to a tribal-area primary school in Jharkhand cannot deliver mother-tongue instruction, because she does not speak the mother tongue. MTB-MLE materials exist; the linguistic capability at the point of delivery does not.

2.2 Decomposition

#	Sub-problem	Consequence today
P1	Teacher cannot read/pronounce the tribal-language lesson script	Falls back to Hindi; MTB-MLE intent lost
P2	Teacher cannot understand what a child says in Santali	Child's response is treated as silence; participation collapses

#	Sub-problem	Consequence today
P3	Teacher cannot produce practice material in the mother tongue	No worksheets; no home practice
P4	No connectivity in the school	Any cloud-dependent tool is unusable
P5	Device is a low-cost tablet	Any large-model solution is unusable
P6	AI output in a low-resource language is unverifiable by the user	Silent propagation of wrong language to children

P6 is the one most competing solutions ignore and the one this PRD treats as a first-class requirement.

2.3 Explicit non-problems

This product does not attempt to teach the teacher Santali, replace the PALASH textbooks, assess children summatively, or serve as a general-purpose translator.

3. Background

3.1 PALASH and the MTB-MLE context

[VERIFIED] PALASH — *Promotion of Appropriate Language and Academic Skills for Holistic Education* — is run by the Jharkhand Education Project Council (JEPC) in collaboration with UNICEF and the Language and Learning Foundation. It has introduced five tribal languages (Kurukh, Santhali, Mundari, Ho, Kharia) into primary education, and as of the January 2026 National Multilingual Education Conclave was operating in eight districts, with the state minister publicly calling for extension to four regional languages (Khortha, Nagpuri, Panchpargania, Kurmali).

Source: [PTI report via Careers360, Jan 2026](#) · [MLE India on PALASH](#)

[VERIFIED] Textbooks in five tribal and two regional languages were developed and NCERT-approved in 2016 and distributed to roughly 1,000 schools; earlier M-TALL/*Bhasha Puliya* work produced bilingual picture dictionaries in nine languages.

Source: [CALA 2019 / SOAS](#)

Product implication: the state already owns validated bilingual print material and a trained-teacher network. Bhasha Setu should ingest that material as its seed corpus rather than translate NCERT Hindi from scratch. **[VALIDATE]** — availability of digital source files from JEPC is unknown to the team and must be requested.

3.2 NIPUN Bharat

[VERIFIED] NIPUN Bharat (launched 5 July 2021 under Samagra Shiksha) targets universal foundational literacy and numeracy by end of Grade 3 by 2026-27. Learning outcomes are organised under three developmental goals — **HW** (Health and Wellbeing), **EC** (Effective Communicators), **IL** (Involved Learners) — divided into six levels spanning three ECCE years plus

Grades 1–3, and **each learning outcome carries an official number/code for referencing**. Targets are published as the *Lakshya Soochi*, prepared by NCERT. Source: [NIPUN Bharat guidelines \(PIB PDF\)](#) · [PIB launch release](#)

The codes exist and are official. This team does not have them. Section 19 therefore specifies a placeholder scheme and the exact source of record that must populate it before any deployment claim is made.

3.3 Language technology landscape

Summarised here, evidenced in full in Section 41.

- **[VERIFIED]** IndicTrans2 (AI4Bharat) covers all 22 Scheduled Languages including **Santali in Ol Chiki (sat_01ck)**; distilled 200M variants are published. Neither Ho nor Mundari is a Scheduled Language and neither is covered.
- **[VERIFIED]** IndicConformer publishes a **Santali (sat)** ASR model (Conformer-Large, 120M-parameter encoder, hybrid CTC-RNNT) alongside a 600M multilingual model, MIT licensed, with ONNX artefacts.
- **[VERIFIED]** Indic Parler-TTS officially supports Santali with named voices, trained on a corpus containing ~298 hours of Santali.
- **[VERIFIED]** Bhashini exposes a multilingual AI4Bharat conformer ASR service that lists Santali; Bhashini is a cloud API and is therefore a *content-pipeline* tool, not an on-device one.
- **[VERIFIED]** Adi Vaani (Ministry of Tribal Affairs, IIT Delhi-led consortium) covers Santali, Bhili, **Mundari**, and Gondi, built on IndicTrans2 and NLLB. **[RESEARCH]** whether any Adi Vaani weights, data, or API are available for third-party use — public reporting describes an app, not an SDK.
- **[VERIFIED]** AdiBhashaa is a community-curated open parallel corpus and MT benchmark for Bhili, Mundari, Gondi and Santali (arXiv 2512.04765); a dataset appears on Hugging Face under `misniitdelhi/AdiBhasha` . **[VALIDATE]** size, license and quality before use.

4. Target Users

User	Count / context	Primary need
Primary teacher (Grades 1–3) , Hindi-medium trained, posted in a tribal-area school	The core user. PS cites 5,000+ schools	Deliver today's lesson in the child's language without knowing it
Children (ages 5–9) , Santali/Mundari/Ho home language	Beneficiary, never a direct operator of the device	Hear instruction and be understood
Native-speaker reviewer (community member, tribal-language teacher, TRI/M-TALL linguist)	Small, high-leverage cohort	Approve or reject AI output before children ever see it
Block/District education officer (BRP/CRP)	Supervisory	Push content packages, see adoption, verify offline

User	Count / context	Primary need
		install base
State content administrator (JEPC)	Very small	Own the curriculum, own the publish button

Explicit non-users for MVP: parents, students as independent learners, private schools.

5. User Personas

Sunita Devi — the teacher (primary persona). 34, B.Ed., posted to a single-teacher primary school in West Singhbhum. Teaches Grades 1–3 simultaneously. Speaks Hindi and Nagpuri; no Santali beyond greetings. Uses WhatsApp fluently, has never installed an app from anywhere but the Play Store. Her phone is her own; the tablet is school property and is shared. She has 4–6 minutes of setup time before a class, no time during it. **Design consequence:** every classroom action must be reachable in ≤2 taps from the dashboard, and nothing may require typing during a lesson.

Birsa Murmu — the reviewer (secondary persona). 46, Santali-speaking government teacher in Dumka, writes Ol Chiki fluently, reviews in evening batches on a phone over a 3G-class connection. Paid per reviewed item or contributing as part of a PALASH resource-group role. **Design consequence:** review must be a queue of small decisions with audio playback, usable on a phone, tolerant of connection loss, and must never require him to type long corrections when a spoken correction would do.

Anjali Kumari — the BRP (tertiary persona). Manages 40 schools, visits each monthly, carries a laptop and a mobile hotspot. Is the realistic *sync vector* for schools with no connectivity at all. **Design consequence:** content packages must be transferable from her device to a tablet without internet (local Wi-Fi Direct / USB / SD card).

6. User Journeys

6.1 Journey A — Teaching a prepared lesson (highest frequency)

1. Sunita opens the tablet in the classroom. App launches to dashboard, offline badge green, "Grade 1 · Santali" retained from last use.
2. Taps **Teach** → today's lesson is already at the top (last-taught + next-in-sequence).
3. Lesson opens split-screen: Hindi teacher script on the left, Santali (Ol Chiki + Devanagari transliteration) on the right, a large ► per line.
4. She reads the Hindi line, taps ►, the tablet speaks the verified Santali line aloud to the class.
5. Children respond. For anything she doesn't understand she taps **Listen** (Live mode inline), a child speaks, she reads the Hindi gist.
6. At the end she taps **Worksheet** → a bilingual worksheet for this lesson's learning outcome is already generated and cached; she prints it later at the block office or copies it to the blackboard.

Success criterion: steps 1–4 complete in under 60 seconds from cold app start. **[Target]**

6.2 Journey B — Unscripted classroom moment

Sunita needs to say something not in the script ("Take out your slate and write the letter"). She holds the mic button, speaks Hindi, releases. Fast path recognises it as phrase-bank entry `CMD-047`, plays the pre-recorded Santali audio. She sees a green **✓ Verified** chip. Total time from release of button to audio start: target <1.5 s.

If the utterance is *not* in the phrase bank, the dynamic path runs, the result is spoken by TTS, and the chip is amber **⚠ AI translation — not verified**. She can tap **Report** to push it into the review queue.

6.3 Journey C — Child speaks

Sunita taps **Child speaks**, hands the tablet toward the child (or leaves it on the desk), the child answers in Santali. Santali ASR → Hindi text appears large on screen. She reads it. Optionally Hindi TTS speaks it into her earpiece. Failure is expected and designed for: if confidence is low, the app shows *"Didn't catch that — ask the child to repeat"* rather than a guess.

6.4 Journey D — Review loop

Birsa opens the reviewer web app, sees 40 pending items, plays the audio, reads the Hindi and the Santali, and taps **✓** / **✗** / **🗨**. Rejections require a reason code; corrections can be typed in Ol Chiki or recorded as audio. Approved items enter the next content package; corrections accumulate as fine-tuning data.

6.5 Journey E — Sync

Anjali arrives at the school, opens **Sync** on both devices, the tablet pulls the delta package over local Wi-Fi Direct, verifies checksums, applies it atomically, and reports "Santali Grade 1 content updated to v14 · 62 MB · 4 new lessons".

7. Product Vision

An offline AI vernacular classroom copilot that lets a Hindi-medium teacher teach foundational literacy and numeracy in the child's mother tongue — with every word the child hears traceable to a human who approved it.

Three-year framing: Bhasha Setu becomes the delivery layer for PALASH content, and the correction stream it generates becomes the largest education-domain parallel corpus for Jharkhand's tribal languages — the product improves the language technology it depends on.

8. Goals

ID	Goal	Measure
G1	Teacher delivers an FLN lesson in Santali without prior language training	Lesson completion rate in-app; observed classroom sessions
G2	Two-way classroom dialogue within 3 s	p50 and p95 end-to-end latency, measured per Section 27
G3	Zero unverified tribal-language content presented as authoritative	100% of <code>PUBLISHED</code> content has a reviewer ID; unverified output always visually marked
G4	Full functionality with radio off	Airplane-mode test suite passes for every P0 feature
G5	Runs on Android 9, ~2 GB RAM	Benchmark plan in Section 27.3 passes on the reference device
G6	Every generated artefact traces to a learning outcome	Schema constraint: worksheets/flashcards require a non-null <code>learning_outcome_id</code>

9. Non-Goals

1. **Not** a general-purpose translator (no news, no medical, no legal domain).
2. **Not** three languages at prototype stage. One, done properly.
3. **Not** a student-facing app; children never operate the device unsupervised.
4. **Not** an assessment/reporting system for the state in MVP.
5. **Not** a teacher-training or language-learning app.
6. **Not** cloud-inference-at-runtime. Cloud exists only for content preparation and sync.
7. **Not** a replacement for native-speaker teachers — explicitly a bridge until they are recruited.
This framing matters politically and should be stated in the pitch.
8. **No** child voice data leaves the device. Ever. (Section 26.)

10. Product Principles

1. **Verified beats fluent.** A blunt approved sentence outranks an elegant generated one.
2. **Never hide uncertainty from the teacher.** Verified/unverified state is always visible; low confidence produces an honest failure, not a guess.
3. **Offline is the default state, not the degraded state.** The online path is the exception.
4. **The classroom is the latency budget.** If it doesn't respond before the children's attention breaks, it failed regardless of quality.
5. **Cheap devices are the target, not the fallback.** Optimise for the 2 GB tablet first; anything better is headroom.

6. **The community owns the language.** Native speakers approve, correct, and are credited. Corrections flow back into the models.
7. **Build for the teacher's hands, not the engineer's demo.** Two taps, big targets, audio-first.
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11. Functional Requirements

Priority: **P0** = required for SIH demo; **P1** = important, post-demo; **P2** = future.

11.1 Content and lessons

ID	Requirement	Pri
FR-1.1	Browse curriculum by Grade → Subject → Topic → Lesson	P0
FR-1.2	Display Hindi and tribal-language lesson script side by side	P0
FR-1.3	Play verified audio per line, sentence-level granularity	P0
FR-1.4	Show Devanagari transliteration of OI Chiki text, toggleable	P0
FR-1.5	Show the learning outcome for every lesson	P0
FR-1.6	Show verification state on every tribal-language string	P0
FR-1.7	Teacher notes / activity instructions / assessment prompts per lesson	P0
FR-1.8	Favourite and "recently taught" lists	P1

11.2 Live classroom

ID	Requirement	Pri
FR-2.1	Push-to-talk Hindi capture with VAD endpointing	P0
FR-2.2	Hindi ASR on device	P0
FR-2.3	Phrase-bank fast-path match with cached audio playback	P0
FR-2.4	Dynamic path: Hindi ASR → MT → tribal TTS	P0
FR-2.5	Reverse path: tribal ASR → Hindi text	P0
FR-2.6	Display recognised Hindi text for teacher confirmation before playback (configurable)	P0
FR-2.7	Visible verified/unverified marking of every spoken output	P0
FR-2.8	On-screen latency readout in demo mode	P0

ID	Requirement	Pri
FR-2.9	Report-a-bad-translation action, queued for review	P1
FR-2.10	Reverse path speaks Hindi via TTS	P1

11.3 Worksheets and flashcards

ID	Requirement	Pri
FR-3.1	Generate bilingual worksheet from (grade, topic, LO, difficulty, item count)	P0
FR-3.2	Worksheet items drawn from verified item bank, assembled deterministically	P0
FR-3.3	Answer key + teacher instructions + LO stamp on every worksheet	P0
FR-3.4	Export to PDF, stored locally, shareable	P0
FR-3.5	Flashcard deck generation: image + Hindi + tribal text + audio + transliteration	P0
FR-3.6	Full-screen flashcard presentation mode with tap-to-hear	P0
FR-3.7	Teacher-authored custom item added to local bank (unverified, marked)	P2

11.4 Offline and sync

ID	Requirement	Pri
FR-4.1	100% of P0 features work with radio off after first sync	P0
FR-4.2	Language pack download/removal with size shown before download	P0
FR-4.3	Manifest-driven delta sync with checksum verification	P0
FR-4.4	Atomic package application with rollback on failure	P0
FR-4.5	Resumable download over intermittent connection	P1
FR-4.6	Peer-to-peer sync (device→device, no internet)	P1
FR-4.7	Storage pressure handling: evict least-used audio, never evict models	P1

11.5 Verification

ID	Requirement	Pri
FR-5.1	Reviewer web app with queue, audio playback, approve/reject/correct	P0 (backend + minimal UI for demo credibility)

ID	Requirement	Pri
FR-5.2	State machine enforced server-side (Section 21)	P0
FR-5.3	Corrections stored as training data with provenance	P1
FR-5.4	Reviewer attribution and contribution stats	P2

12. Core Features

12.1 Teach Mode (P0)

The default surface. Selection state (grade, language) is sticky. The lesson view is a two-column script with per-line playback, an activity block, a materials block, and an assessment block, each carrying its own translations and audio. All content is `PUBLISHED` state only — Teach Mode never shows machine output.

Key design decision: Teach Mode contains **no live model inference at all**. It is a fast, reliable content reader. This guarantees that the most-used feature is also the most robust and the least battery-hungry, and it means Teach Mode still works if every model fails to load.

12.2 Live Classroom Mode (P0)

The 3-second feature. Full architecture in Section 16. Two visible states — **Teacher → Class** (Hindi in, Santali out) and **Child → Teacher** (Santali in, Hindi out) — toggled by a single large switch, because mode confusion is the most likely usability failure in a live classroom.

12.3 Worksheet Generator (P0)

This is deliberately not an LLM feature. Worksheets are assembled from a verified, LO-tagged item bank using templates (match-the-following, fill-in-the-blank, count-and-write, picture-word matching, read-and-circle). Deterministic assembly means every printed worksheet is composed entirely of human-approved language, it runs instantly on a 2 GB device, and it is reproducible. Generative composition of *new* items is a cloud-side, review-gated pipeline (Section 20), not an on-device one.

Output: A4 PDF, high-contrast, printer-and-photocopier friendly (pure black on white, no greys under 20%, no full-bleed images), Hindi and tribal language in parallel columns, LO code in the footer.

12.4 Flashcard Generator (P0)

Decks by category (fruits, animals, numbers 1–20, letters/OI Chiki characters, colours, shapes, body parts, classroom objects, family). Each card: image (locally bundled, openly licensed), Hindi word, tribal word in native script, transliteration, audio button. Presentation mode is full-screen, swipeable, one tap to hear.

Image sourcing [VALIDATE]: must be culturally appropriate to Jharkhand (a "house" card should not show a suburban American house). Source from openly-licensed collections and/or commission simple line art; every image asset must carry a license record in the content package. This is a real work item, not a footnote.

12.5 Content verification console (P0 backend, P1 polish)

Web app for reviewers. See Section 21.

12.6 Sync and downloads (P0)

See Section 17.

13. UX / UI Requirements

13.1 Constraints that drive the design

The user has 6 minutes before class and none during it; may have never used a tablet before this posting; cannot read the tribal script; is standing, holding the device in one hand, in a room with 30 children.

13.2 Global rules

- **Two taps to anything.** Dashboard → feature → action.
- **Minimum touch target 56 dp;** primary classroom actions ≥72 dp.
- **Body text ≥18 sp, lesson script ≥22 sp,** user-scalable to 200% without layout breakage.
- **Every action has an icon and a Hindi label.** Icon-only is banned; Hindi-only is discouraged.
- **No typing in the classroom path.** Typing exists only in Settings, search, and the reviewer app.
- **Contrast ≥ 7:1** for text (targets WCAG AAA) — devices are used near windows in bright light.
- **OI Chiki rendering must not depend on the system font.** Bundle a licensed OI Chiki font (e.g. Noto Sans OI Chiki, SIL OFL) in the APK. **[VERIFIED]** OI Chiki occupies Unicode U+1C50–U+1C7F; Android 9 system font coverage is not guaranteed → **[VALIDATE]** on the reference device.
- **Every tribal-language string carries a state chip:** green ✓ Verified / amber △ AI / grey ○ Draft. Non-negotiable.

13.3 Screens

#	Screen	Content	Notes
1	Onboarding	4 slides, audio-narrated in Hindi, skippable	No account creation; school code + teacher PIN only
2	Language selection	Big cards: Santali (available) / Mundari, Ho (marked "coming soon", honest)	Shows pack size before download

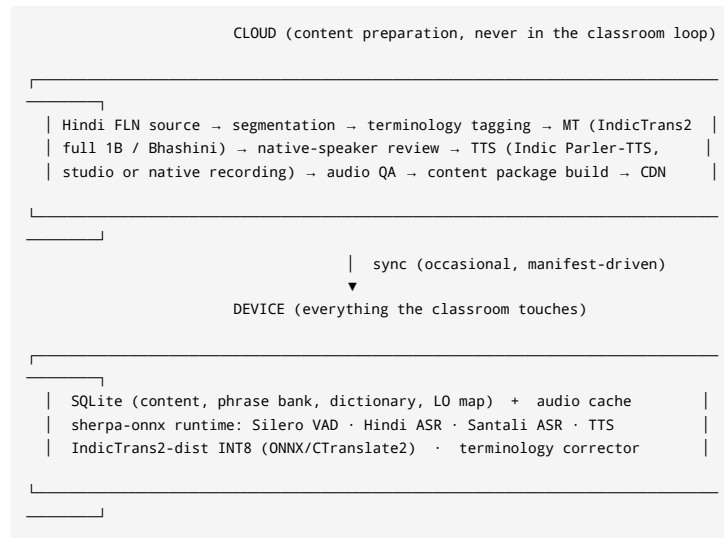
#	Screen	Content	Notes
3	Teacher dashboard	4 large tiles: Teach, Live Class, Worksheets, Flashcards + offline/storage badge	Sticky grade+language selector at top
4	Teach Mode	Grade→Subject→Topic→Lesson drill-down, then split script view	Per-line ►; sticky LO banner
5	Live Classroom	Giant mic button, direction toggle, transcript pane, latency chip	One-handed reachable; mic at bottom
6	Translation result	Hindi text, tribal text, replay, verification chip, Report	Auto-dismiss after 8 s unless pinned
7	Worksheet Generator	Form: grade/subject/LO/difficulty/count → Preview → PDF	Defaults pre-filled from current lesson
8	Flashcard Generator	Category grid → deck → present	Presentation mode hides all chrome
9	Lesson library	Search + filter by LO, offline-available filter	Search is Hindi-only
10	Downloads / sync	Packs, sizes, versions, last sync, Sync now, P2P sync	Progress persists across app kill
11	Content verification	Teacher-side: my reported items and their status	Reviewer console is a separate web app
12	Settings	Language, font size, playback speed, confirm-before-speak, storage, about	Playback speed matters: teachers slow audio for Grade 1
13	Offline status	Persistent badge; detail sheet: models loaded, packs, free space, last sync	Doubles as the demo's credibility screen

13.4 Accessibility (Section requirement)

- Audio-first: every screen's primary instruction is available as spoken Hindi via a speaker icon.
- Low-literacy support: icons + Hindi + audio on all primary actions.
- Large-font mode does not truncate; all layouts tested at 200% scale.
- High-contrast theme; no information conveyed by colour alone (chips carry icon + text).
- Minimal typing; numeric PIN instead of password.
- All assets local; zero network dependency for any UI state.
- Playback speed control (0.75× / 1.0×) for early-grade listening.

14. AI/ML Architecture

14.1 Overview



Rule: no model that produces content children will *repeatedly* hear runs only on-device. On-device models serve the live, transient, teacher-supervised path. Durable content is cloud-generated and human-approved.

14.2 Model inventory

Role	Candidate	Params	Deplc
VAD	Silero VAD (via sherpa-onnx)	~2 M	Devic
Hindi ASR	ai4bharat/indicconformer_stt_hi_hybrid_ctc_rnnt_large	120 M encoder	Devic (ONN INT8)
Santali ASR	IndicConformer sat monolingual	120 M encoder	Devic (ONN INT8)
MT Hi→sat	ai4bharat/indictrans2-en-indic-dist-200M family / indic-indic-dist-320M	200– 320 M	Devic

Role	Candidate	Params	Deplc
MT sat→Hi	<code>indictrans2-indic-en-dist-200M</code> + pivot, or indic-indic 320M	200–320 M	Devic
Tribal TTS	Gap. Indic Parler-TTS supports Santali but is Parler-Mini-class (~880 M)	—	Cloud conte device
Hindi TTS	Piper/VITS Hindi via sherpa-onnx	~30–60 M	Devic
Phrase matcher	SQLite FTS5 + normalised-string + character-trigram fallback	n/a	Devic
Intent/context classifier	Rule + keyword + small logistic model over lesson context	<1 M	Devic

14.3 Per-model specification

Hindi ASR — Input: 16 kHz mono PCM, VAD-segmented, ≤10 s. Output: Hindi Devanagari text + confidence + word timings. Decoding: **CTC greedy** (not RNNT) for the fast path — lower latency, lower memory, adequate for short commands. **[ASSUMPTION]** CTC-greedy is sufficient for classroom command recognition; RNNT reserved for P1. Fallback: if load fails, Live Mode degrades to a browsable phrase-bank grid (teacher taps the phrase instead of saying it) — this fallback alone satisfies the demo's core promise and must be built.

MT — Input: normalised Hindi sentence ≤30 tokens with entity masking as IndicTrans2's `IndicProcessor` requires. Output: OI Chiki text. Decoding: **greedy (beam=1)**, `max_length` capped at 64. Fallback chain: verified phrase bank → dictionary-assisted gloss with explicit "partial translation" marking → refuse with "I can't translate that reliably — please rephrase." **Refusal is a designed output, not a bug.**

TTS — Input: tribal text + voice ID. Output: 22.05 kHz PCM. Primary strategy is **cached audio**, synthesis is the fallback (this inversion is deliberate; see 16.4).

Santali ASR — Input as above. Output: OI Chiki text → MT → Hindi. **[VALIDATE]** IndicConformer's Santali training data almost certainly contains no child speech and no Jharkhand-classroom acoustics. Expect high WER. The UI must be built around that expectation (Section 30).

14.4 The critical on-device datapoint

[VERIFIED, third-party] A published INT8 ONNX export of `indictrans2-indic-en-dist-200M` reports **~5.9 s per short sentence at beam 5, with 109 MB resident**, measured on a Motorola

Edge 40 (Dimensity 8020) — a device considerably more powerful than the 2 GB target tablet. The publisher explicitly notes it is a third-party conversion not verified against official weights. Source: [Hugging Face model card](#)

Interpretation — this is the most important number in the document. A naive on-device neural translation call does not fit inside 3 seconds. It does not fit inside 5. Therefore:

- 1. The fast path (phrase bank + cached audio) is not an optimisation, it is **the primary mechanism** by which the 3-second requirement is met.
- 2. The dynamic path must use **beam=1**, short sequences, INT8, an int-quantised encoder cache, and possibly CTranslate2 rather than ONNX Runtime.
- 3. **[VALIDATE]** measure our own build; beam 5→1 alone typically removes most of the cost, but we have not measured it and must not claim otherwise.
- 4. If validation fails, the honest fallback is: dynamic path is *asynchronous* — the teacher sees "translating..." and the result arrives when it arrives — while all latency-critical classroom speech goes through the phrase bank. The 3-second guarantee is then scoped, explicitly and visibly, to phrase-bank utterances. **State this in the pitch rather than being caught by it.**

15. Translation Architecture

15.1 Language selection: the evidence

Capability	Santali	Mundari	Ho
Scheduled Language (⇒ inside AI4Bharat/Bhashini scope)	Yes	No	No
Open MT model	Yes — IndicTrans2 sat_01ck [VERIFIED]	Adi Vaani beta covers it; weights availability [RESEARCH]	None found
Open ASR model	Yes — IndicConformer sat [VERIFIED]	None found	None found
Open TTS support	Yes — Indic Parler-TTS Santali voices, ~298 h in corpus [VERIFIED]	None found	None found
Open parallel corpus	AdiBhashaa (Santali) [VERIFIED, unvetted]	AdiBhashaa (Mundari) [VERIFIED, unvetted]	None found

Capability	Santali	Mundari	Ho
Script	OI Chiki (U+1C50–U+1C7F), official	Devanagari commonly; no single official standard [VALIDATE]	Warang Citi / Devanagari [VALIDATE]
Speakers (2011 census)	~7.6 M	~1.1 M (approx., verify)	~1.4 M (approx., verify)
Verdict	MVP language	Phase 2	Phase 3

Recommendation: Option A — Hindi ↔ Santali. It is the only choice where all three required capabilities have published open models today, which is precisely what the prototype must demonstrate. Choosing Mundari or Ho would mean building the acoustic and parallel corpora from zero inside a hackathon timeline — the honest outcome of which is a phrase-bank-only demo with no real ASR or MT.

Important caveat to state in the pitch [VALIDATE]: PALASH textbooks in Jharkhand may render Santali in Devanagari rather than OI Chiki, and IndicTrans2 outputs OI Chiki. The product therefore stores **both scripts plus Latin transliteration** for every Santali string and lets the school choose the display script. Confirm the operative script with JEPC before deployment. Do not assume.

15.2 Why generic MT is insufficient here

Three failure classes that generic MT cannot fix:

- **Pedagogical register.** "बच्चों, ध्यान से सुनो" must land as a warm classroom instruction, not a literal command. MT has no notion of register for a 6-year-old.
- **Curriculum terminology.** Numbers, letters, shape names, and the specific vocabulary of the PALASH textbooks must match what the printed book says, or the child sees one word and hears another.
- **Silent hallucination.** In a low-resource pair the model will produce fluent, wrong Santali, and the Hindi-medium teacher cannot detect it. Nothing in the model signals this.

15.3 The hybrid pipeline

```
Hindi input
|
|→ [1] Normalisation: whitespace, Indic digits→ASCII, punctuation, entity
masking (IndicProcessor)
|
|→ [2] Educational context extraction: current grade, subject, topic, LO,
lesson
|      (free – the app already knows what is on screen)
|
|→ [3] EXACT MATCH against verified store (phrase bank + lesson lines)
|      HIT → return verified string + cached audio.  ✓ VERIFIED.  [FAST
PATH ENDS HERE]
|
|→ [4] FUZZY MATCH (normalised edit distance + trigram, threshold τ)
```

```

| HIT → verified string, marked "= matched phrase"
|
|→ [5] Terminology retrieval: pull curriculum glossary entries for tokens
present
|   in the input, scoped to (grade, subject)
|
|→ [6] MT model (IndicTrans2 dist, greedy) with masked glossary terms
|
|→ [7] Terminology correction: unmask and substitute approved tribal terms
|   for every glossary hit – this is a hard override of the model
|
|→ [8] Validation gate:
|   · script check (output is in expected Unicode block)
|   · length-ratio sanity check vs source
|   · OOV check against the tribal dictionary → flag unknown tokens
|   · repetition/degenerate-output detector
|   · numeral consistency check (numbers in = numbers out)
|   FAIL → refuse, show "cannot translate reliably"
|
|→ [9] Output marked △ AI, spoken via TTS, logged for review

```

How this reduces hallucination, concretely: steps 3–4 mean the highest-frequency classroom language never touches the model at all. Step 7 makes curriculum terms deterministic regardless of what the model produced. Step 8 catches the specific, detectable failure signatures of low-resource NMT (wrong script, degenerate repetition, dropped numerals, length blow-up). What remains — a fluent but semantically wrong sentence — is surfaced as *unverified* rather than presented as truth, and is routed to a human. The system never claims certainty it does not have.

15.4 Retrieval design

Phrase bank and glossary live in SQLite with FTS5 over a normalised Hindi column plus a trigram index. **No vector database on device for MVP.** FAISS or embedding retrieval adds an embedding model (memory) and index (storage) for marginal gain over lexical matching on a corpus of a few thousand short, formulaic classroom phrases. **[ASSUMPTION]** — revisit at P2 if fuzzy-match recall proves inadequate in classroom testing.

15.5 Phrase bank construction

Target **800–1,200 classroom utterances** for MVP, sourced by: (a) extracting imperative/interrogative sentences from PALASH teacher handbooks; (b) a structured elicitation session with practising primary teachers ("what do you say in the first ten minutes of class?"); (c) NIPUN activity instructions. Categories: greetings, attention, instructions, praise, correction, materials, transitions, questions, closure. Every entry gets Hindi + Santali (reviewed) + audio (recorded or synthesised then approved) + category + LO tags where applicable.

This is the highest-leverage manual work in the entire project. Start it in Week 1.

16. Speech Architecture

16.1 Two paths

```

FAST PATH (target: covers the majority of classroom utterances)
mic → VAD → Hindi ASR (CTC greedy) → normalise → phrase lookup → cached
Opus/WAV → play

DYNAMIC PATH (tail)
mic → VAD → Hindi ASR → MT (greedy, INT8) → validation → TTS → play

```

16.2 Latency budget — engineering targets, not measurements

Budget is measured from **end of speech (VAD endpoint)** to **first audio sample played**. This definition matters and must be stated on the demo screen; measuring from "start of speech" makes the number depend on how long the teacher talks.

Stage	Fast path target	Dynamic path target
VAD endpoint decision	200 ms	200 ms
ASR (≤4 s utterance, CTC greedy)	400 ms	400 ms
Normalisation	10 ms	10 ms
Phrase lookup (SQLite FTS5)	30 ms	30 ms (miss)
Terminology retrieval	—	40 ms
MT (greedy, ≤30 tokens)	—	900 ms
Validation gate	—	20 ms
TTS synthesis (first chunk)	—	700 ms
Audio decode + playback start	120 ms	120 ms
Framework/JNI overhead + UI	150 ms	200 ms
Total target (p50)	~0.9 s	~2.6 s
Target p95	1.5 s	3.0 s (at the limit)

These are budgets to design against and then measure. Section 14.4 shows the MT figure is the one at risk. Every stage above has an assigned owner and a tracked measurement in the benchmark harness.

16.3 Measurement methodology (must be built in Week 2, not Week 8)

- Instrument each stage with `SystemClock.elapsedRealtimeNanos()` at defined boundaries; emit a structured trace per utterance.
- Fixed test set: **100 recorded Hindi classroom utterances** (50 phrase-bank, 50 out-of-bank), recorded in a real classroom and in a quiet room, by ≥4 speakers.
- Run on the **reference device** (Section 27.3), airplane mode, battery >50%, screen on, after 5-minute warm-up and after a 30-minute soak (thermal state matters).

- Report p50/p90/p95/p99 per stage and end-to-end, plus RSS and CPU%. Publish the harness in the repo — **a demo that shows its own latency distribution is far more credible than one that claims a number.**

16.4 The TTS gap — stated plainly

[VERIFIED] Indic Parler-TTS officially supports Santali with named voices and its pretraining corpus contains ~298 h of Santali. **[VERIFIED]** It is a Parler-Mini-class model, which is far beyond what a 2 GB Android tablet can run in real time. **[RESEARCH]** No public small-footprint (VITS/Piper-class) Santali voice was found.

Three strategies, in the order the team should pursue them:

1. **Cached audio first (P0, guaranteed to work).** Every string in the verified content store and phrase bank ships with pre-generated audio in the content package (produced in the cloud with Indic Parler-TTS and/or recorded by native speakers, then QA'd by a reviewer). Opus at ~24 kbps mono; **[ASSUMPTION]** ~3 KB/s $\Rightarrow$ 1,000 phrases $\times$ 4 s $\approx$ 12 MB. This alone delivers Teach Mode and the fast path completely.
2. **Train a small Santali VITS voice (P1, the real fix).** Fine-tune a VITS/Piper-class model on Santali audio generated by Indic Parler-TTS and/or native recordings, export to ONNX, run via sherpa-onnx. **[VERIFIED]** sherpa-onnx runs VITS/Piper TTS on Android via prebuilt AARs under Apache-2.0. **[VALIDATE]** whether the resulting voice is intelligible enough for Grade 1 phonics — synthetic-teacher-audio for *literacy instruction* is a genuinely higher bar than for general TTS, since children imitate pronunciation.
3. **Native-speaker recording at scale (P2, the correct long-term answer).** For a state deployment, the audio a child hears should be a human voice from their own community. Treat synthesis as scaffolding, not the destination. This is also the strongest ethical and political framing for the pitch.

16.5 On-device runtime choice

sherpa-onnx (k2-fsa) is the recommended speech runtime: **[VERIFIED]** Apache-2.0, wraps ONNX Runtime, ships prebuilt Android AARs for arm64-v8a/armeabi-v7a/x86/x86_64, supports streaming and non-streaming ASR, VITS/Piper TTS, Silero VAD, and has published NeMo model export scripts (relevant because IndicConformer is a NeMo model) plus Dart/Flutter bindings on pub.dev. This single dependency covers VAD + ASR + TTS with one native layer, which materially reduces integration risk.

Sources: [sherpa-onnx](#) · [Flutter package](#) · [pretrained models](#)

16.6 Reverse path (child $\rightarrow$ teacher)

Same runtime, Santali ASR model, output routed through MT to Hindi. Because expected WER is high (child speech, dialectal variation, no matched training data), the UI contract is: show the Hindi *and* the recognised Santali; if ASR confidence is below threshold, do not translate — show "Didn't catch that." Never fabricate a child's answer. **A wrong transcription of a child's answer is worse than no transcription**, because the teacher may mark the child wrong.

17. Offline-First Architecture

17.1 Pipeline

```
Cloud: content preparation → MT → native validation → audio generation → QA
      → package build (manifest + shards + checksums) → object storage/CDN
              |
              sync (HTTP delta, or P2P from a BRP device)
              ▼
Device: staging dir → checksum verify → atomic apply → SQLite + audio store
      → app reads only from local store, always, online or not
```

17.2 Package structure

```
santali-grade1-v14/
manifest.json      # version, deps, shard list, sha256 per shard, total
size
content.sqlite.gz # lessons, translations, worksheets items, flashcards,
LOs
phrasebank.sqlite.gz # classroom phrases + audio index
audio/             # sharded opus, ~10 MB per shard
images/           # flashcard/worksheet art + license records
fonts/            # Ol Chiki (only in the base pack)
models/           # referenced by ID, downloaded separately, shared
across packs
```

17.3 Manifest (illustrative)

```
{
  "package_id": "sat-g1-content",
  "version": 14,
  "min_app_version": "1.2.0",
  "language": "sat",
  "grade": 1,
  "depends_on": [{"package_id": "sat-base", "min_version": 3}],
  "total_bytes": 64821004,
  "shards": [
    {"path": "content.sqlite.gz", "bytes": 8412233, "sha256": "..."},
    {"path": "audio/shard-000.tar", "bytes": 10485760, "sha256": "..."}
  ],
  "delta_from": {"13": ["content.sqlite.gz", "audio/shard-004.tar"]},
  "created_at": "2026-01-15T00:00:00Z",
  "signature": "ed25519:..."
}
```

17.4 Protocol rules

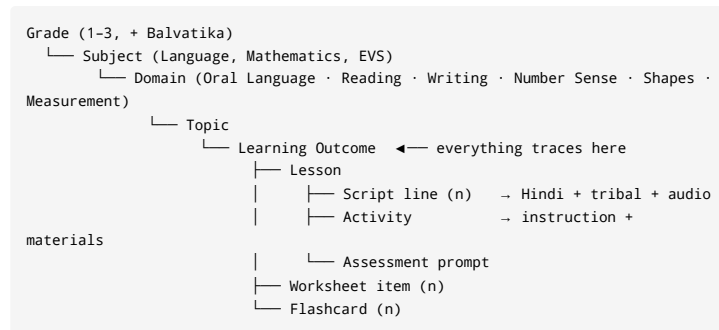
- **Versioning:** monotonic integer per package; packages are immutable once published. Models are versioned separately and referenced by ID, so a content update never re-downloads a 200 MB model.
- **Delta updates:** manifest declares which shards changed from version *N*; client downloads only those.

- **Atomicity:** download to `staging/`, verify every sha256, then apply inside a single SQLite transaction plus a directory swap. Power loss at any point leaves the previous version intact.
- **Corruption recovery:** checksum failure → discard shard, retry ×3 with backoff → mark package degraded → app continues on the previous good version and shows a repair action.
A corrupt package must never brick the classroom.
- **Conflict handling:** content is server-authoritative and read-only on device; there is no content merge problem by design. Device→server data is append-only (reported items, anonymous usage counters, review corrections) with client-generated UUIDs, making retries idempotent.
- **Partial updates:** each shard is independently applicable; a partially-synced package is usable at its previous version and resumes at the shard boundary.
- **Storage limits:** hard budget in Section 27.2. On pressure: evict audio for non-current grades first, then images, then optional language packs. **Never evict models or the current grade's content;** instead block new downloads and tell the teacher what to remove.
- **P2P sync (P1):** BRP device advertises packages over Wi-Fi Direct/local hotspot; same manifest and checksum path, plus signature verification (ed25519 public key pinned in the app) so a compromised intermediate device cannot inject content.

17.5 What lives on the tablet

Lessons, translations (all scripts), audio, worksheet item bank, generated worksheet PDFs, flashcard decks + images, tribal dictionary, phrase bank, curriculum + LO map, models, config, and a local event log. Full inventory with size budget in Section 27.2.

18. Curriculum Architecture



Hard invariant, enforced by a NOT NULL foreign key: no lesson, worksheet item, flashcard, or assessment prompt may exist without a `learning_outcome_id`. If the team cannot say which outcome a piece of content serves, the content does not enter the system. This single constraint is what makes "NIPUN-aligned" a checkable property rather than a slide claim.

19. NIPUN / FLN Alignment

19.1 What is verified

[VERIFIED] Three developmental goals — HW (Health and Wellbeing), EC (Effective Communicators), IL (Involved Learners); key competencies under each; one or more learning outcomes per competency; **each learning outcome carries an official number/code**; six levels spanning Balvatika→Grade 3; targets published as the *Lakshya Soochi* by NCERT. ([NIPUN Bharat Guidelines, PIB](#))

19.2 What is NOT in this document

The actual outcome codes. They are official and must be transcribed from the source, not reconstructed. Any code appearing in the prototype must use the placeholder namespace below, and the UI must render placeholders in grey with the label "code pending official mapping".

Placeholder scheme: PLACEHOLDER : : <GOAL> - <GRADE> - <SEQ> — e.g. PLACEHOLDER : : EC-G1-014 .

Source of record to populate them (do this before any state conversation):

1. NIPUN Bharat Guidelines, Department of School Education & Literacy, MoE — the goal/competency/LO tables.
2. NCERT *Lakshya Soochi* (FLN targets, Balvatika–Grade 3).
3. Jharkhand's state-adapted FLN targets via JEPC — states are expected to adapt national targets, so **the Jharkhand adaptation, not the national list, is the operative mapping.**

19.3 Example mapping (structure real, code placeholder)

Layer	Content
Learning outcome	PLACEHOLDER : : EC-G1-014 — "Reads simple words with familiar letters" (<i>illustrative wording; replace with official text</i>)
Lesson	"ᱵᱚᱠᱟ ᱦᱚᱱᱚᱛ letters — first five sounds" (Grade 1 · Language · Reading)
Teacher instruction	Hindi: "बच्चों, इस अक्षर की आवाज़ सुनो" → verified Santali + audio
Activity	Sound-hunt: children find classroom objects starting with the sound; bilingual object names from flashcard deck classroom-objects
Worksheet	8 items: 4 letter-sound matching, 4 picture-word matching, bilingual, answer key, LO code in footer
Assessment prompt	"Show me the letter that makes this sound" — Hindi + verified Santali + audio
Flashcards	Deck ol-chiki-letters-1 , 5 cards, image + Hindi + Santali + audio

Every row above is queryable by LO code, which is what makes the "show NIPUN alignment" demo step (Section 39, step 13) a live database query rather than a static slide.

20. Data Strategy

20.1 Content pipeline

```
Hindi FLN source (PALASH handbooks / NCERT FLN / state material)
  → ingestion + provenance record (source, page, license)
  → sentence segmentation + cleanup
  → curriculum tagging (grade, subject, domain, topic, LO, content_type,
difficulty)
  → terminology extraction → curriculum glossary (Hindi term → approved tribal
term)
  → MT (IndicTrans2 full model / Bhashini, cloud – quality over speed here)
  → terminology enforcement pass
  → NATIVE-SPEAKER REVIEW ← mandatory gate, cannot be skipped or automated
  → approved tribal text (Ol Chiki + Devanagari + Latin transliteration)
  → audio generation (Indic Parler-TTS cloud, or native recording)
  → AUDIO QA (native listener: intelligibility, pronunciation, pacing for Grade
1)
  → content package build → sign → publish
```

20.2 Core content schema

```
content_item(
  content_id          TEXT PRIMARY KEY,      -- uuid
  content_type        TEXT NOT NULL,         -- lesson_line | activity |
assessment
                                                    -- | worksheet_item | flashcard |
phrase | glossary
  grade              INTEGER NOT NULL,
  subject            TEXT NOT NULL,
  domain             TEXT,
  topic              TEXT,
  learning_outcome_id TEXT NOT NULL REFERENCES learning_outcome(lo_id),
  hindi_text         TEXT NOT NULL,
  language           TEXT NOT NULL,         -- ISO 639-3: sat | mxx | unr |
hoc
  tribal_text        TEXT,                  -- native script
  tribal_text_deva   TEXT,                  -- Devanagari rendering
  transliteration     TEXT,                  -- Latin
  audio_path         TEXT,
  audio_source       TEXT,                  -- tts_model_id |
native_recording
  difficulty         INTEGER,               -- 1..3
  verification_state TEXT NOT NULL,        --
DRAFT|AI_GENERATED|REVIEW_REQUIRED
                                                    --
|NATIVE_VERIFIED|PUBLISHED|REJECTED
  verified           INTEGER NOT NULL DEFAULT 0,
  reviewer_id        TEXT REFERENCES reviewer(reviewer_id),
  reviewed_at        TEXT,
  mt_model_version   TEXT,                  -- provenance: which model
produced it
  version            INTEGER NOT NULL DEFAULT 1,
```

```

source_ref      TEXT,          -- book/page/license provenance
created_at      TEXT NOT NULL,
updated_at      TEXT NOT NULL
);

```

`verified = 1` **only** when `verification_state = 'PUBLISHED'`, enforced by a CHECK constraint and by the server. The device treats `verified` as the sole authority for showing a green chip.

20.3 The correction flywheel

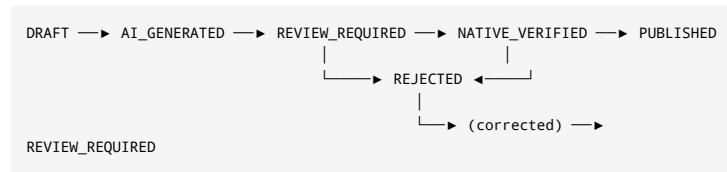
Every reviewer correction is stored as a triple: `(hindi_source, mt_output, human_correction, model_version, reviewer_id, timestamp, reason_code)`. This produces:

- **Immediately:** new verified phrase-bank entries and glossary terms — the fast path grows with use.
- **Per release:** a curated fine-tuning set for IndicTrans2 in the education domain (the exact domain gap that public models lack).
- **Long term:** an openly publishable Hindi–Santali education-domain parallel corpus, which is arguably a more durable contribution than the app. Publish it back to the community (aligning with the AdiBhashaa/Bhasha Daan model) with reviewer credit.

[ASSUMPTION] correction volume from a small reviewer pool will be modest (hundreds, not tens of thousands) in year one; plan fine-tuning around low-data adaptation (LoRA/adaptor on the distilled model), not full retraining.

21. Native Speaker Validation System

21.1 State machine



Enforced rules:

- Only `PUBLISHED` content ships in a content package.
- Only `PUBLISHED` content displays a green ✓ and is used in Teach Mode, worksheets, and flashcards.
- Transition to `NATIVE_VERIFIED` requires a `reviewer_id` and a timestamp. Server-side, non-bypassable.
- `PUBLISHED` requires `NATIVE_VERIFIED` **plus** audio QA pass for any item with audio.
- Live dynamic-path output is **never** written into the content store as anything but `AI_GENERATED`.

21.2 What reviewers check

Translation adequacy · grammar and morphology · curriculum terminology match · audio pronunciation and pacing · script correctness (Ol Chiki orthography) · **educational and cultural appropriateness for a 6-year-old** (a technically correct translation can still be wrong for a classroom).

21.3 Reviewer workflow design

Queue of atomic items; each shows Hindi, proposed tribal text, audio, and context (grade/LO/lesson). Actions: ✓ Approve · ✗ Reject (reason code required) · ✎ Correct (type or record). Two-reviewer agreement required for items entering the phrase bank (highest reuse, highest blast radius); single review sufficient for individual worksheet items. **[ASSUMPTION]** — tune after measuring inter-reviewer disagreement.

Reviewer recruitment is a real dependency, not a technical one: PALASH resource persons, tribal-language teachers, TRI/M-TALL linguists, and Santali-medium college students. Named in the risk register (Section 38).

21.4 The non-negotiable

The system must never silently treat AI-generated tribal-language content as authoritative.

Implementation: verification state is a required field on every tribal string in every API response and every UI component; the mobile app fails closed (renders the amber chip) if the field is missing. There is no code path that displays tribal text without a state chip.

22. Backend Architecture

22.1 Component map

Component	Cloud	On-device	Notes
API gateway	✓	—	FastAPI app; rate limiting, routing
Auth service	✓	partial	JWT issue in cloud; offline PIN + cached token on device
Curriculum service	✓	✓ (read-only SQLite)	Source of truth in cloud
Translation service (batch, high quality)	✓	—	Full IndicTrans2 / Bhashini; content pipeline only
Translation (live)	—	✓	Distilled INT8 model
Speech service (ASR/TTS, batch)	✓	—	Audio generation for packages
Speech (live)	—	✓	sherpa-onnx

Component	Cloud	On-device	Notes
Content generation (worksheet/flashcard authoring)	✓	—	LLM-assisted item drafting → review gate
Worksheet assembly	—	✓	Deterministic, from verified item bank
Flashcard assembly	—	✓	Deterministic
Sync service	✓	✓ client	Manifest + shard serving
Model registry	✓	✓ manifest cache	Model IDs, versions, checksums, target ABI
Content registry	✓	—	Package index and publish state
Review service	✓	—	State machine + queue
Admin dashboard	✓	—	JEPC content ops
Telemetry	✓	✓ buffer	Anonymous, aggregate, opt-in (Section 26)

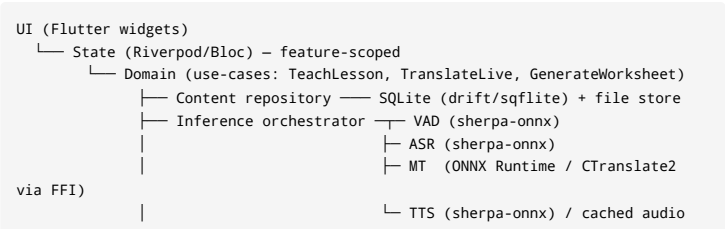
The classroom loop touches zero cloud components. That is the architectural claim being made, and it is testable by putting the device in airplane mode — which is exactly what the demo does.

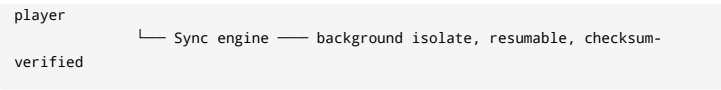
22.2 Deployment shape

Single FastAPI monolith with clean internal module boundaries, Postgres, object storage (S3-compatible/MinIO), Redis for job queueing, Docker Compose for the hackathon and a single VM/managed container service for pilot. **No Kubernetes, no microservices, no event bus for MVP.** The cloud side has approximately ten concurrent users (content team + reviewers); operational complexity is pure cost here.

23. Mobile Architecture

23.1 Layers





23.2 Memory and lifecycle policy (the part that decides whether 2 GB works)

Model	Residency policy
VAD	Always resident when Live Mode is open (tiny)
Hindi ASR	Loaded on entering Live Mode, unloaded on exit or after 60 s idle
MT	Lazy: loaded on first fast-path miss; unloaded after 120 s idle
Santali ASR	Loaded only when direction = Child→Teacher; mutually exclusive with...
TTS	...loaded only when a dynamic-path synthesis is required

Rule: never hold ASR + MT + TTS simultaneously in the default flow. The pipeline is sequential, so the orchestrator loads stage *N+1* while releasing stage *N* wherever the latency budget allows, and holds at most two models at once. **[VALIDATE]** — load/unload cost may itself violate the latency budget, in which case the fallback is to keep ASR + MT resident and rely on cached audio exclusively for output (dropping on-device TTS from the MVP). That trade is acceptable; the reverse (keeping TTS and losing ASR) is not.

Inference runs in a background isolate; all model handles are explicitly disposed;
`onTrimMemory` / `onLowMemory` triggers immediate unload of everything except VAD and the content DB.

23.3 Flutter vs native Kotlin

Recommendation: Flutter. **[VERIFIED]** sherpa-onnx publishes maintained Dart/Flutter packages with prebuilt Android native libraries, which removes the main historical argument for going native (speech JNI plumbing). Flutter gives one codebase for tablet + phone, a much faster UI iteration loop for a hackathon timeline, and an admin/reviewer web build from the same code if desired.

Honest counterpoint: native Kotlin gives finer control over memory, thermal behaviour, and ONNX Runtime threading — all of which matter on a 2 GB device — and avoids the Flutter engine's own footprint. **[VALIDATE]** measure Flutter baseline RSS on the reference device in Week 1. If the Flutter engine plus UI exceeds ~180 MB resident, revisit this decision immediately rather than at Week 7.

24. Database Schema

24.1 Cloud (PostgreSQL) — abbreviated DDL

```
CREATE TABLE school (
  school_id UUID PRIMARY KEY, udise_code TEXT UNIQUE, name TEXT NOT NULL,
```

```

district TEXT, block TEXT, default_language TEXT REFERENCES language(code));

CREATE TABLE app_user (
  user_id UUID PRIMARY KEY, role TEXT NOT NULL,          --
  teacher|reviewer|brp|admin
  name TEXT NOT NULL, phone TEXT UNIQUE, pin_hash TEXT NOT NULL,
  school_id UUID REFERENCES school(school_id), created_at TIMESTAMPTZ DEFAULT
  now());

CREATE TABLE language (
  code TEXT PRIMARY KEY,                                -- sat, mјx, unr, hoc,
  hi
  name TEXT, native_name TEXT, primary_script TEXT,      -- Olck, Deva, ...
  alt_scripts TEXT[], is_active BOOLEAN DEFAULT false);

CREATE TABLE learning_outcome (
  lo_id TEXT PRIMARY KEY,                                -- PLACEHOLDER::EC-G1-
  014
  official_code TEXT,                                    -- NULL until
  populated from source
  goal TEXT CHECK (goal IN ('HW','EC','IL')),
  competency TEXT, grade INT, subject TEXT,
  description_hi TEXT, description_en TEXT,
  source_document TEXT, is_placeholder BOOLEAN DEFAULT true);

CREATE TABLE lesson (
  lesson_id UUID PRIMARY KEY, grade INT, subject TEXT, topic TEXT,
  lo_id TEXT NOT NULL REFERENCES learning_outcome(lo_id),
  title_hi TEXT, sequence INT, duration_minutes INT);

CREATE TABLE translation (
  translation_id UUID PRIMARY KEY, content_id UUID NOT NULL,
  language TEXT REFERENCES language(code),
  text_native TEXT, text_deva TEXT, transliteration TEXT,
  mt_model_version TEXT, verification_state TEXT NOT NULL,
  reviewer_id UUID REFERENCES app_user(user_id), reviewed_at TIMESTAMPTZ,
  version INT DEFAULT 1,
  UNIQUE (content_id, language, version));

CREATE TABLE audio_asset (
  audio_id UUID PRIMARY KEY, translation_id UUID REFERENCES
  translation(translation_id),
  storage_key TEXT, duration_ms INT, codec TEXT, sample_rate INT,
  source TEXT, qa_state TEXT, qa_reviewer_id UUID);

CREATE TABLE phrase_bank (
  phrase_id TEXT PRIMARY KEY,                            -- CMD-047
  category TEXT, hindi_text TEXT NOT NULL, hindi_normalised TEXT NOT NULL,
  language TEXT, tribal_text TEXT, audio_id UUID REFERENCES
  audio_asset(audio_id),
  verification_state TEXT NOT NULL, usage_count BIGINT DEFAULT 0);

CREATE TABLE worksheet_item (
  item_id UUID PRIMARY KEY, lo_id TEXT NOT NULL REFERENCES
  learning_outcome(lo_id),
  template TEXT, difficulty INT, payload JSONB, verification_state TEXT NOT

```

```

NULL);

CREATE TABLE flashcard (
  card_id UUID PRIMARY KEY, deck TEXT, image_key TEXT, image_license TEXT,
  hindi_word TEXT, lo_id TEXT REFERENCES learning_outcome(lo_id));

CREATE TABLE model_version (
  model_id TEXT, version TEXT, task TEXT,          -- asr|mt|tts|vad
  language TEXT, artifact_key TEXT, bytes BIGINT, sha256 TEXT,
  quantisation TEXT, min_app_version TEXT, PRIMARY KEY (model_id, version));

CREATE TABLE sync_package (
  package_id TEXT, version INT, language TEXT, grade INT,
  manifest JSONB, total_bytes BIGINT, published_at TIMESTAMPTZ,
  PRIMARY KEY (package_id, version));

CREATE TABLE verification_record (
  record_id UUID PRIMARY KEY, translation_id UUID, reviewer_id UUID,
  action TEXT,                                     --
  approve|reject|correct
  reason_code TEXT, correction_text TEXT, correction_audio_key TEXT,
  created_at TIMESTAMPTZ DEFAULT now());

```

24.2 Device (SQLite) — read-optimised subset

```

CREATE TABLE content (
  content_id TEXT PRIMARY KEY, content_type TEXT, grade INTEGER, subject TEXT,
  topic TEXT, lo_id TEXT NOT NULL, hindi_text TEXT NOT NULL,
  tribal_text TEXT, tribal_text_deva TEXT, transliteration TEXT,
  audio_path TEXT, verified INTEGER NOT NULL, difficulty INTEGER,
  lesson_id TEXT, sequence INTEGER, pkg_version INTEGER);

CREATE TABLE phrase (
  phrase_id TEXT PRIMARY KEY, category TEXT,
  hindi_normalised TEXT NOT NULL, tribal_text TEXT NOT NULL,
  audio_path TEXT NOT NULL, verified INTEGER NOT NULL DEFAULT 1);
CREATE VIRTUAL TABLE phrase_fts USING fts5(hindi_normalised, content='phrase',
                                             content_rowid='rowid',
                                             tokenize='unicode61');

CREATE INDEX idx_content_lo ON content(lo_id);
CREATE INDEX idx_content_lesson ON content(lesson_id, sequence);

CREATE TABLE glossary (
  term_hi TEXT, grade INTEGER, subject TEXT, term_tribal TEXT,
  PRIMARY KEY (term_hi, grade, subject));

CREATE TABLE dictionary (word_tribal TEXT PRIMARY KEY, gloss_hi TEXT, pos TEXT);

CREATE TABLE local_event (
  -- append-only, anonymous,
  uploaded on sync
  event_id TEXT PRIMARY KEY, type TEXT, payload TEXT, created_at TEXT, synced
  INTEGER DEFAULT 0);

CREATE TABLE reported_item (
  report_id TEXT PRIMARY KEY, hindi_text TEXT, mt_output TEXT,

```

```
lo_id TEXT, created_at TEXT, synced INTEGER DEFAULT 0);

CREATE TABLE pkg_state (package_id TEXT PRIMARY KEY, version INTEGER, applied_at TEXT, health TEXT);
CREATE TABLE model_state (model_id TEXT PRIMARY KEY, version TEXT, path TEXT, sha256 TEXT, bytes INTEGER);
```

Note: device tables carry no student names, no teacher-identifying free text, and no audio recordings. That is a schema-level privacy guarantee, not a policy promise.

25. API Specification

All endpoints are prefixed `/api/v1`. Auth is `Authorization: Bearer <JWT>` unless noted. Errors follow a single envelope:

```
{"error": {"code": "TRANSLATION_UNAVAILABLE", "message": "...", "retryable": true}}
```

Endpoint	Purpose	Auth	Offline behaviour
POST /auth/login	school code + PIN → JWT (+ 90-day offline token)	none	Cached token permits full offline use until expiry
POST /translate	Hindi → tribal, high quality (content pipeline)	teacher/admin	Device does not call this. On-device pipeline is used instead
POST /speech/transcribe	audio → text	admin	Not used at runtime; pipeline/QA only
POST /speech/translate	audio → translated audio	admin	Not used at runtime
POST /tts	text → audio (package build)	admin	Not used at runtime
POST /worksheet/generate	LLM-assisted item drafting → REVIEW_REQUIRED	admin	Device assembles locally from verified bank

Endpoint	Purpose	Auth	Offline behaviour
POST /flashcard/generate	deck drafting	admin	Device assembles locally
GET /lessons	curriculum tree	teacher	Served from local SQLite
GET /content-packages	available packages for (lang, grade)	teacher	Cached list; sync button disabled offline
GET /content-packages/{id}/manifest	manifest + shard checksums	teacher	—
GET /content-packages/{id}/shard/{n}	shard bytes (range requests supported)	teacher	Resumable
POST /sync	upload append-only events + reports; get delta list	teacher	Queued locally, flushed on connectivity
GET /model-manifest	models for ABI + app version	teacher	Cached
GET /review/queue	reviewer's pending items	reviewer	Web only
POST /review/{id}/decision	approve/reject/correct	reviewer	Web only

Representative contracts:

```

POST /api/v1/translate
{ "text": "सब बच्चे किलाब खोलो", "source_lang": "hi", "target_lang": "sat",
  "context": { "grade": 1, "subject": "language", "lo_id": "PLACEHOLDER::EC-G1-014"},
  "require_verified": true }

200 OK
{ "translation_id": "...", "text_native": "...", "text_deva": "...",
  "transliteration": "...", "verification_state": "PUBLISHED",
  "match_type": "phrase_bank_exact", "phrase_id": "CMD-047",
  "audio_url": "...", "confidence": null, "unknown_tokens": [] }

409 TRANSLATION_UNVERIFIED // when require_verified=true and only AI output exists
503 TRANSLATION_UNAVAILABLE // model down; client falls back to phrase bank

```

```
POST /api/v1/sync
{ "device_id": "...", "packages": [{ "package_id": "sat-g1-content", "version": 13 }],
  "events": [{ "event_id": "...", "type": "lesson_completed", "payload": { ... } }],
  "reports": [{ "report_id": "...", "hindi_text": "...", "mt_output": "... } }

200 OK
{ "accepted_events": 42, "accepted_reports": 3,
  "updates": [{ "package_id": "sat-g1-content", "latest_version": 14,
    "delta_shards": [ "content.sqlite.gz", "audio/shard-004.tar" ],
    "bytes": 12488212 } ] }
```

Idempotency: all client-generated IDs are UUIDs; the server upserts on primary key, so retries after connection loss are safe.

26. Security and Privacy

Children are the subjects here. The default answer to every data question is "don't collect it."

Concern	Decision
Child voice recordings	Never persisted, never transmitted. Audio buffers live in memory only, are overwritten immediately after inference, and are not written to disk even temporarily. No "improve our models with your recordings" toggle in MVP
Teacher voice	Same — transient buffers only
Student PII	Not collected. No names, no roll numbers, no enrolment IDs, no photos
Teacher PII	Name + phone + school, minimum viable for account recovery; stored encrypted at rest
Local storage	Content DB is not sensitive; auth token and PIN hash in Android Keystore-backed EncryptedSharedPreferences; app-private storage only
Transport	TLS 1.2+, certificate pinning for the sync host
Package integrity	ed25519 signature verified before apply; public key pinned in the APK
Consent	School/DEO-level deployment consent + teacher in-app consent for anonymous usage analytics; children are never asked to consent to anything because nothing about them is collected
Retention	Events: 12 months aggregate. Reported translations: retained (they contain no personal data — Hindi utterance + MT output only, with a scrub pass for anything name-like)
Analytics	Opt-in, aggregate counters only (lessons opened, translations attempted, fast-path hit rate, latency histograms, crash traces). No content of live speech, no per-child anything

Concern	Decision
Device sharing	Tablets are shared; PIN unlock, no per-child state, nothing personal on screen after lock
Offline processing	All student speech processed on-device by construction

Applicable frameworks to review before deployment [VALIDATE]: India's DPDP Act 2023 (children's data provisions — verifiable parental consent, prohibition on tracking/behavioural monitoring of children and on targeted advertising) and any Jharkhand education department data policy. The architecture above is designed to sit comfortably inside the strictest reading, by collecting nothing about children at all.

27. Performance Requirements

27.1 Reference device

[ASSUMPTION] Reference target: Android 9 (API 28), 2 GB RAM, 4× ARM Cortex-A53-class @ ~1.8 GHz, arm64-v8a, 16–32 GB storage, no NPU/GPU compute assumed. Procure at least one such physical device in Week 1. **Emulator benchmarks are not acceptable evidence** for any claim in this document.

27.2 Budgets (targets)

Resource	Budget	Rationale
APK size	≤ 60 MB	Play Store / sideload over weak connections; models are downloaded, not bundled
Peak RSS during live translation	≤ 550 MB	Leaves headroom under Android's low-memory killer on a 2 GB device with ~1.1–1.3 GB available [VALIDATE]
Idle RSS (Teach Mode)	≤ 220 MB	Teach Mode holds no models
Model storage (Santali pack)	≤ 450 MB	ASR-hi ~130 MB + ASR-sat ~130 MB + MT ~110 MB + TTS ~60 MB, all INT8 [VALIDATE] — these are extrapolations from parameter counts, not measured file sizes
Content + audio (Grade 1)	≤ 150 MB	~1,000 phrases + lesson audio in Opus
Total install footprint	≤ 800 MB	Fits a 16 GB tablet with other apps present

Resource	Budget	Rationale
Cold start → dashboard	≤ 2.5 s	
Teach Mode lesson open	≤ 500 ms	Pure DB read
Fast-path end-to-end (p95)	≤ 1.5 s	
Dynamic path end-to-end (p95)	≤ 3.0 s	The PS requirement. At risk — see 14.4
Battery	≥ 4 h continuous mixed classroom use	[VALIDATE]
Sustained thermal	No throttling-induced latency regression >20% after 30 min	[VALIDATE]

27.3 Benchmark plan (a gate, not a report)

- 1. **Week 2:** instrumentation harness lands with the first ASR integration. Every build emits stage timings and peak RSS for a fixed 100-utterance corpus.
- 2. **Week 4:** first full-pipeline measurement on the reference device. **Gate: if dynamic-path p95 > 4 s, the team formally descopes on-device dynamic TTS and pivots to cached-audio-only output.** Decide this in Week 4, not Week 8.
- 3. **Week 7:** soak test — 30 minutes continuous Live Mode, thermal + battery + memory curve.
- 4. **Every release:** the same harness runs and results are committed to `/docs/benchmarks/` with device, build, and date. **The repo should contain the raw numbers, including bad ones.** Judges and, later, a state procurement process trust measured mediocrity over claimed excellence.

Until step 2 completes, this project does not claim it runs on 2 GB RAM. It claims it is designed to, and states the plan for proving it.

28. Evaluation Framework

Layer	Metric	Method	Target
Translation	BLEU, chrF++	Held-out Hindi–Santali education-domain test set (200–500 sentences, native-authored)	Report absolute values; [RESEARCH] no defensible target exists for this pair — establish our own baseline

Layer	Metric	Method	Target
	Human adequacy (1–5)	2 native reviewers, blind	≥4.0 mean on published content
	Human fluency (1–5)	Same	≥4.0
	Terminology accuracy	% of glossary terms rendered with the approved term	100% (deterministic by construction — measure to prove the override works)
	COMET	If a Santali-capable evaluator model is available [RESEARCH]	—
ASR	WER / CER (Hindi, adult, classroom noise)	100-utterance corpus	Report; target WER ≤20% for phrase-bank matching to work
	WER / CER (Santali, child)	Small child-speech corpus recorded with consent from teachers, not children [VALIDATE] feasibility	Report honestly; expect poor
	Fast-path hit rate	% of live utterances matched to phrase bank	≥70% is the design assumption [VALIDATE]
Speech translation	End-to-end latency p50/p95	Section 16.3 harness	p95 ≤3 s
	End-to-end intelligibility	Native listener rates whether the played audio conveys the intent (1–5)	≥4.0
System	Peak RSS, CPU%, battery/hour, storage, cold start, offline success rate	Automated harness + airplane-mode suite	Section 27.2
Education	Teacher SUS score	Standard 10-item SUS after 3 sessions	≥70
	Task completion rate	Can the teacher complete Journey A unaided?	≥90%
	Lesson completion	In-app counter	Baseline first
	Child comprehension	Teacher-administered oral check on lesson	[RESEARCH] — a real study, not a hackathon

Layer	Metric	Method	Target
		vocabulary	claim

On educational impact: a hackathon prototype cannot demonstrate learning gains. Say so. Propose the study design (a term-long comparison in PALASH schools with and without the tool, measuring against FLN targets) as future work rather than implying results you do not have. Judges reward this; they penalise fabricated impact numbers.

29. KPIs

Product (post-pilot): weekly active teachers; lessons taught per teacher per week; Live Mode sessions per lesson; fast-path hit rate; worksheets generated and printed; % of devices on the latest content version; median days between syncs.

Quality: CRTS (below); reported-translation rate per 1,000 utterances; review queue latency (submission → decision); % content in `PUBLISHED` .

Technical: dynamic-path p95 latency; crash-free session rate ≥99%; sync success rate; package corruption rate; peak RSS distribution across the fleet.

29.1 Proposed product metric

CRTS — Classroom-Ready Translation Score
A proposed internal product metric, not an established academic measure. It is not comparable to BLEU, chrF, or any published benchmark and must never be presented as one.

For a sampled utterance i , three native-speaker judgements on 0–1 scales:

- A_{*i*} — Translation adequacy:** does it mean what the Hindi meant?
- P_{*i*} — Pedagogical appropriateness:** is it right for this grade, this outcome, this register, this culture? (A grammatically perfect sentence a 6-year-old can't parse scores low.)
- T_{*i*} — Terminology conformance:** do curriculum terms match the PALASH/state-approved vocabulary?

$$\text{CRTS} = \frac{1}{N} \sum_{i=1}^N \left(w_A A_i \cdot w_P P_i \cdot w_T T_i \right)^{1/(w_A + w_P + w_T)}$$

with default weights **w_A = 0.4, w_P = 0.35, w_T = 0.25**. The multiplicative (weighted geometric) form is deliberate: a zero on any dimension zeroes the score. A pedagogically wrong translation is not "70% good" — it is unusable, and an additive score would hide that.

Reporting rules: sample ≥100 utterances per release, stratified across fast path and dynamic path, scored by ≥2 native reviewers with inter-rater agreement reported alongside. Report fast-path and dynamic-path CRTS separately — they will differ substantially, and averaging them conceals exactly the thing the metric exists to expose.

30. Failure Modes

#	Failure	Likelihood	Mitigation
F1	Hindi ASR misrecognises in classroom noise	High	VAD + push-to-talk (not always-on); show recognised text before speaking (toggle default ON); one-tap retry; near-miss fuzzy phrase match
F2	Teacher's regional accent / Nagpuri-influenced Hindi	High	Phrase bank matches on normalised text + trigrams, tolerating ASR error; collect misrecognitions for future fine-tuning
F3	Code-switching (Hindi + Nagpuri + English)	High	Glossary covers common English classroom loans; unmatched tokens flagged as unknown rather than translated
F4	Multiple children speaking at once	Certain	Reverse path is explicitly single-speaker; UI instructs "one child at a time"; VAD rejects overlapping/long segments with "too noisy — try again"
F5	Unknown word (out of dictionary)	High	Validation gate flags OOV tokens; UI shows them highlighted with "not in dictionary"; never silently guessed
F6	Incorrect translation, fluent output	Medium	Verified-first architecture; amber chip; Report button; reviewer loop; teacher trained (in onboarding) that amber means "use with care"
F7	Hallucinated tribal words	Medium	Terminology override + OOV check + repetition detector + length-ratio check; refuse rather than emit
F8	TTS mispronunciation (esp. Ol Chiki phonics)	High	Cached, human-QA'd audio is the primary path; synthesis limited to non-phonics utterances; audio QA gate blocks bad pronunciation from ever shipping
F9	Device overheats / thermal throttling	Medium	Sequential model residency; unload after idle; soak test; degrade to fast-path-only when <code>PowerManager</code> reports thermal status $\geq$ MODERATE
F10	Low battery	High (rural power)	Battery-saver mode disables dynamic path, keeps Teach Mode + cached audio; warn at 15%
F11	Storage exhaustion	High (16 GB shared tablets)	Pre-download size check; eviction policy (audio for other grades $\rightarrow$ images $\rightarrow$ optional packs); never evict models or the current grade

#	Failure	Likelihood	Mitigation
F12	Corrupted package	Medium	Per-shard sha256; atomic apply; rollback to last good version; repair action
F13	Model fails to load / crashes	Medium	Try-catch around every load; on failure the feature degrades (Live → tappable phrase grid; dynamic → phrase-bank only) and reports; the app never crashes because a model did
F14	No network for months	Certain	Designed for; content simply ages. Warn at >90 days: "content may be outdated"
F15	ASR silently transcribes a child's answer wrongly	High	Confidence threshold → "Didn't catch that"; always show the recognised tribal text alongside the Hindi so a bilingual aide can catch errors
F16	Teacher over-trusts the tool	Medium	Persistent verification chips; onboarding explicitly teaches the amber state; Teach Mode (verified-only) is the default surface, Live Mode is secondary

31. MVP Scope (SIH Prototype)

One language: Hindi ↔ Santali. **One grade:** Grade 1. **One subject pair:** Language + Numeracy basics. Depth over breadth.

In scope (P0):

1. Teach Mode: ~10 complete Grade 1 lessons, verified Santali text (OI Chiki + Devanagari + transliteration), cached audio per line, LO stamp.
2. Live Classroom: push-to-talk, Hindi ASR on device, phrase bank of 800+ verified utterances with cached audio, dynamic fallback path, verification chips, on-screen latency readout.
3. Reverse path: Santali ASR → Hindi text (demonstrated honestly, including its failure behaviour).
4. Worksheet Generator: deterministic assembly from a verified item bank, bilingual PDF with answer key and LO code.
5. Flashcard Generator: 4 decks (numbers 1–10, animals, fruits, classroom objects) with images and audio.
6. Offline: full airplane-mode operation; language pack download; manifest + checksum sync.
7. Verification: state machine enforced end-to-end; a working reviewer console (can be minimal) showing real items moving DRAFT → PUBLISHED.
8. Benchmark harness with published numbers.

Explicitly NOT in the MVP — do not build these:

Mundari and Ho support · user accounts beyond school code + PIN · student profiles or progress

tracking · analytics dashboards for officials · on-device model fine-tuning · P2P sync · handwriting or OCR input · generative image creation for flashcards · multi-grade content beyond Grade 1 · text chat or LLM tutor · gamification · a parent app · cloud live translation.

Every hour spent on the second list is an hour not spent making the first list survive a live demo.

32. Phase 2 (Pilot, ~3–6 months)

Mundari via AdiBhashaa/Adi Vaani collaboration · Grades 2–3 content · trained small Santali VITS voice on device · P2P sync from BRP devices · reviewer console at scale with two-reviewer agreement and payments · fine-tuned IndicTrans2 education-domain adapter from correction data · teacher-authored custom phrases (unverified, marked) · offline usage analytics + district dashboard · classroom pilot study design with JEPC · accessibility audit · Hindi TTS for the reverse path.

33. Phase 3 (Scale)

Ho and the remaining PALASH languages, contingent on corpus creation with TRIs · native-speaker audio recording programme replacing synthesis · integration with state systems (UDISE+/Vidya Samiksha Kendra) **[VALIDATE]** feasibility · open publication of the education-domain parallel corpus and voice datasets · offline device fleet management · Balvatika/ECCE content · a Santali literacy assessment aligned to the state-adapted Lakshya Soochi · sustainability model (state procurement vs. grant funding) · third-party security and privacy audit before any statewide rollout.

34. Technical Stack

Layer	Choice	Why
Mobile	Flutter (Dart)	One codebase; [VERIFIED] maintained <code>sherpa_onnx</code> Flutter package with prebuilt Android natives removes the speech-JNI problem that usually forces native; fast UI iteration. Re-evaluate against Kotlin if Week-1 RSS measurement is poor (§23.3)
Speech runtime	sherpa-onnx (Apache-2.0, over ONNX Runtime)	VAD + ASR + TTS in one dependency; NeMo export scripts exist for IndicConformer; prebuilt AARs for all four ABIs
MT runtime	ONNX Runtime Mobile , with CTranslate2 as the benchmark alternative	ORT keeps one runtime for everything; CTranslate2 is purpose-built for encoder-decoder MT with INT8 on CPU and may win on latency. Measure both in Week 3, pick on data

Layer	Choice	Why
Local DB	SQLite (drift) + FTS5	Ships with Android; FTS5 gives phrase lookup with no extra dependency; no vector DB needed at this corpus size
Backend	FastAPI (Python)	Same language as the ML tooling (NeMo, transformers, IndicTransToolkit); async I/O suits sync/shard serving; team velocity
Cloud DB	PostgreSQL	Relational content model with hard FK constraints — which is the mechanism enforcing "every artefact has a learning outcome"; JSONB for item payloads
Object storage	S3-compatible (MinIO in dev)	Shard and audio hosting with range requests for resumable download
Queue	Redis + RQ/Celery	Batch TTS and MT jobs in the content pipeline
ML training	PyTorch + NeMo + HF transformers	Native format of every candidate model
Retrieval	SQLite FTS5 + trigram ; FAISS only server-side if needed	Avoids an embedding model on a 2 GB device (§15.4)
PDF	ReportLab / WeasyPrint server-side; pdf package on device	Device generation must be offline-capable
Deployment	Docker Compose → single VM	Ten users on the cloud side; Kubernetes is pure overhead
CI	GitHub Actions	Build APK, run tests, run the benchmark harness on an emulator for regressions (device runs are manual and recorded)
Monitoring	Sentry (self-hosted or free tier) + structured logs + Prometheus if pilot-scale	Lightweight; crash reporting matters more than metrics at this stage

Licence discipline (important, often fatal in government contexts):

[VERIFIED] IndicConformer is MIT; Indic Parler-TTS is Apache-2.0 (verify on the current model card); sherpa-onnx Apache-2.0; ONNX Runtime MIT. **[VERIFIED]** NLLB-200 is **CC-BY-NC-4.0** — **non-commercial only**, which makes it unsuitable for a government deployment even though Adi Vaani reportedly uses it. Maintain `/docs/licenses.md` recording every model, dataset, font, and image asset with its licence and source URL, and verify each one against the live model card before submission. Do not inherit a licence claim from a blog post.

35. Repository Structure

```

bhasha-setu/
├── apps/
│   ├── mobile/                # Flutter app – teacher-facing, all offline
├── logic
│   ├── lib/features/{teach, live, worksheet, flashcard, sync, settings}/
│   ├── lib/core/{db, inference, audio, sync, telemetry}/
│   ├── android/               # ABI config, ProGuard, native libs
│   └── admin/                  # Reviewer console + content ops (web)
├── services/
│   └── api/                    # FastAPI gateway, auth, curriculum, sync
├── endpoints
│   ├── translation/           # Batch MT for the content pipeline
│   ├── speech/                # Batch ASR/TTS, audio QA tooling
│   └── content/                # Package builder, manifest signer, publisher
├── ml/
│   └── translation/            # IndicTrans2 export, quantisation, eval
├── harness
│   ├── asr/                   # IndicConformer → ONNX export, INT8, eval
│   └── tts/                    # Santali voice experiments, VITS training
├── scripts
│   └── evaluation/             # BLEU/chrF/WER/CRTS scoring, report generation
├── data/
│   └── curriculum/             # Hindi FLN source + LO map (+ provenance
├── records)
│   ├── language/              # Glossary, dictionary, transliteration tables
│   └── phrase-bank/            # The 800-1,200 classroom utterances + audio
├── index
│   └── licenses/               # Per-asset licence records
├── bench/                      # Latency/memory harness + committed results
├── docs/                       # This PRD, ADRs, API spec, benchmark reports,
├── licences
├── tests/                      # Unit, integration, offline (airplane-mode)
├── suites
└── infra/                      # Docker Compose, migrations, deploy scripts

```

Directory	Owner responsibility
apps/mobile	All device behaviour: offline-first correctness, model lifecycle, latency instrumentation, verification chips
apps/admin	Review state machine UI; content publishing
services/*	Cloud content preparation and sync only — never in the classroom path
ml/*	Model export, quantisation, evaluation. Owns the claim that a model fits on the device
data/*	The verified corpus. Highest long-term value asset in the repo
bench/	The credibility of every performance claim in this document

Directory	Owner responsibility
<code>docs/</code>	ADRs recording <i>why</i> — especially the Flutter, ORT-vs-CTranslate2, and TTS decisions

36. Implementation Roadmap (8 weeks)

Sequenced so the riskiest unknowns are hit early and the demo path is working end-to-end by Week 5, leaving three weeks to make it survive contact with reality.

Week 1 — Foundation + procure the truth

Repo, CI, Flutter shell, SQLite schema, FastAPI skeleton. **Buy the 2 GB Android 9 tablet.** Measure Flutter baseline RSS on it. Start phrase-bank elicitation with teachers (this is a long-lead item). Request PALASH digital source material and the Jharkhand FLN target list from JEPC.

Week 2 — Speech first, not translation

Integrate sherpa-onnx: VAD + Hindi ASR on device. **Ship the latency harness in the same week.** Record the 100-utterance test corpus. First measured ASR numbers on the reference device. *Why speech before translation:* the ASR integration is the highest-uncertainty piece with the longest debugging tail, and the harness must exist before any optimisation work begins.

Week 3 — Fast path + MT reality check

Phrase bank in SQLite with FTS5; exact + fuzzy matching; cached audio playback. **Fast path end-to-end and measured.** In parallel: export IndicTrans2-dist to ONNX and CTranslate2, quantise INT8, benchmark both on the device against the §14.4 datapoint. **Decision point: which MT runtime, and does the dynamic path survive?**

Week 4 — Curriculum + Teach Mode

Curriculum schema loaded with 10 real Grade 1 lessons; content pipeline for MT + review; Teach Mode UI with per-line audio. First reviewer session with actual native speakers on real content. **Gate: benchmark review (§27.3) — descope dynamic TTS now if the numbers demand it.**

Week 5 — Offline + sync

Package builder, manifest, checksums, atomic apply, rollback, language pack download. **Full airplane-mode test suite green.** This is the week the product becomes the product it claims to be.

Week 6 — Worksheets + flashcards

Verified item bank, deterministic assembly, PDF generation, 4 flashcard decks with licensed images and audio. Reverse path (Santali ASR → Hindi) integrated with honest failure UI.

Week 7 — Optimisation + hardening

Model residency policy, load/unload tuning, thermal and soak testing, battery profiling, memory-pressure handling, crash-proofing every model load, failure-mode drills (pull the storage, corrupt a package, kill the model).

Week 8 — Testing, demo, submission

Teacher usability session (real teacher, unaided, SUS + task completion), full benchmark run published to `/docs/benchmarks/`, demo rehearsal on the reference device ×10, demo video, README, licence manifest, GitHub repository finalised.

Standing rule: at the end of every week the app must install and run a complete lesson offline on the reference tablet. If it doesn't, that is the only thing worked on the following Monday.

37. Testing Strategy

Layer	Coverage
Unit	Normalisation, fuzzy matcher, validation gate (each check independently), worksheet assembly determinism, manifest parsing, eviction policy
Integration	Full fast path with a recorded audio fixture; dynamic path with mocked models; sync apply + rollback; package corruption injection
Offline suite	Every P0 user journey executed with the radio off, as an automated instrumentation test. A failure here blocks release
Device matrix	Reference 2 GB Android 9 tablet (primary); one mid-range Android 12 phone; one Android 14 device — for API-level regressions, not for performance claims
Performance	Section 16.3 harness on every build; results committed
Resilience drills	Kill process mid-sync; fill storage to <100 MB; corrupt a shard; delete a model file; airplane mode mid-translation; low battery; 30-minute thermal soak
Linguistic QA	Native reviewer sign-off on 100% of shipped content; CRTS sample per release
Usability	≥3 teachers, unaided, Journey A and B, think-aloud, SUS
Accessibility	200% font scale, contrast checks, audio-only navigation of primary flows
Security	Dependency scan, no secrets in the APK, signature verification test with a deliberately bad signature

38. Risk Register

P = probability, I = impact (1–5).

ID	Risk	P	I	Mitigation	Owner
R1	No on-device Santali TTS exists ; synthesis path unachievable in timeline	4	5	Cached audio is the primary path by design (§16.4); dynamic TTS is P1, not P0. Demo does not depend on it	ML
R2	On-device MT exceeds latency	4	5	Fast path carries the 3 s claim; greedy decoding; CTranslate2 evaluation;	ML

ID	Risk	P	I	Mitigation	Owner
	budget (§14.4 evidence)			Week-4 gate with a pre-agreed descope	
R3	Santali ASR quality on child speech is unusable	4	3	Reverse path presented with honest confidence handling; not the headline feature; teacher-mediated	ML
R4	Hindi-Santali education-domain parallel data is scarce	5	3	Verified-first architecture reduces dependence on MT quality; AdiBhashaa [VALIDATE] ; correction flywheel builds the corpus we need	Data
R5	Native-speaker reviewers not recruited in time	3	5	Start Week 1; PALASH resource persons, tribal-language teachers, TRI contacts; MVP needs a handful, not hundreds. Without reviewers there is no verified content and the product's core claim collapses	Product
R6	2 GB RAM proves insufficient with models resident	3	4	Sequential residency; unload policy; Teach Mode requires no models; Week-1 baseline measurement; worst case ships as phrase-bank + content reader, which still solves P1/P3	Mobile
R7	Model sizes exceed storage on 16 GB shared tablets	3	3	INT8 everything; language packs; eviction policy; measured sizes published	Mobile
R8	Licence contamination (e.g. NLLB CC-BY-NC in a government deployment)	3	5	<code>/docs/licenses.md</code> verified against live model cards; NLLB excluded; legal review before pilot	Eng lead
R9	Wrong script assumption (OI Chiki vs Devanagari in Jharkhand classrooms)	3	4	Store all three renderings; school-level display setting; confirm with JEPC	Product
R10	Official NIPUN codes unavailable before submission	4	2	Placeholder namespace, visibly marked, with the source of record named (§19.2). Honesty here is a strength, not a gap	Product
R11	Child privacy failure (recording, retention, or a	2	5	Collect nothing about children; no audio persistence anywhere in the	Eng lead

ID	Risk	P	I	Mitigation	Owner
	regulator's reading of DPDP)			code path; schema-level guarantee; pre-pilot audit	
R12	Government deployment friction (procurement, device supply, MDM)	4	3	Out of scope for MVP; design for sideload + P2P sync so the tool is useful without a state contract	Product
R13	Teacher over-trusts unverified output and teaches wrong language	3	5	Verification chips everywhere; Teach Mode verified-only by default; onboarding covers it explicitly; report loop	Product/UX
R14	Demo fails live (device, audio, thermal)	3	4	Rehearse ×10 on the actual device; airplane mode throughout; recorded backup video; no live network dependency anywhere	Team

39. SIH Demo Flow (4 minutes)

Run entirely on the reference tablet, screen-mirrored, **in airplane mode from the very start** — do not turn the internet off mid-demo as a reveal; start with it off and point at the status bar. It is a stronger claim and removes a failure mode.

#	Time	Action	On screen	Say
1	0:00	Frame the problem	Photo: tribal-area classroom	"5,000+ schools. The teacher speaks Hindi. The child doesn't."
2	0:20	Show airplane mode + offline badge	Status bar + green offline chip	"Everything after this point is on this tablet. No network."
3	0:30	Dashboard → Grade 1 → Santali	Dashboard	"Two taps to a lesson."
4	0:45	Open FLN lesson	Split Hindi / Ol Chiki script, LO banner	"Every line traces to a NIPUN learning outcome."
5	1:00	Tap ► on a line	Audio plays; green ✓ Verified chip	"That audio was approved by a native speaker. Not generated on the fly."

#	Time	Action	On screen	Say
6	1:15	Enter Live Classroom	Big mic, direction toggle	
7	1:25	Speak Hindi: "सब बच्चे किताब खोलो"	Recognised Hindi → Santali text → audio plays; latency chip shows measured ms	"Phrase bank hit. Under a second, measured on this device."
8	1:45	Speak something off-script	Amber △ AI chip, latency chip	"Now the model runs. Marked unverified — the teacher always knows which is which."
9	2:05	Child speaks Santali (teammate)	Santali text + Hindi gist	"And back the other way."
10	2:20	Generate worksheet	Form → bilingual PDF with answer key + LO code	"Assembled from approved items. Instant, offline, printable."
11	2:40	Flashcard deck	Full-screen card, tap to hear	
12	2:55	LO traceability	Filter by learning outcome → lesson + worksheet + flashcards appear	"This is a live query, not a slide."
13	3:10	Verification console	Item moving REVIEW_REQUIRED → PUBLISHED	"No AI Santali reaches a child without a human approving it."
14	3:25	Architecture + benchmarks	One diagram; the committed benchmark table with p50/p95	"Here are our real numbers, including where we're weakest."
15	3:45	Close	Roadmap: Santali → Mundari → Ho	"One language, done properly, extensible by design."

Demo discipline: show the latency measurement rather than asserting it; show one amber unverified result deliberately (it demonstrates the safety architecture and pre-empt the judge's hardest question); never claim Mundari or Ho work.

40. Competitive Differentiation

Capability	Google Translate	Generic LLM	Bhashini	Generic edu apps	Adi Vaani	Bhasha Setu
Santali support	No	Unreliable	Yes (cloud)	No	Yes	Yes
Mundari / Ho	No	No	No	No	Mundari yes	Roadmap, stated honestly
Works fully offline	Partial (packs)	No	No	Varies	Offline mode reported [VALIDATE]	Yes, by design
Curriculum-aware (grade/topic/LO)	No	No	No	Sometimes	No	Yes
NIPUN/FLN outcome traceability	No	No	No	Rare	No	Yes, enforced by schema
Native-speaker verification gate	No	No	Model-level, not per string	No	Community validation reported	Yes, per string, visible in UI
Classroom workflow (script, activity, assessment)	No	No	No	Partial	No	Yes
Bilingual worksheet + flashcard generation	No	Text only	No	Yes (not tribal)	No	Yes
Real-time two-way classroom speech	Partial	No	Yes (cloud, needs network)	No	Speech reported	Yes, on-device
Runs on 2 GB Android 9	Partial	No	Thin client only	Varies	[VALIDATE]	Design target, being measured

The honest positioning: Bhashini and Adi Vaani are *language infrastructure* and this project should say so publicly and build on them rather than claim to beat them. What does not exist is the

classroom layer — curriculum-bound, outcome-traceable, verification-gated, and functional with the radio off. That is the contribution.

Claims not made: that our translation quality beats Bhashini's or Adi Vaani's (untested, and unlikely for general text); that we support three languages; that we have any government partnership; that we have measured educational impact.

41. Open-Source Resources (evidence table)

Resource	Open	Languages relevant here	Licence	Size	Offline	2 GB feasible
IndicTrans2	Yes	All 22 scheduled incl. Santali sat_01ck ; not Ho/Mundari	MIT (verify per artefact)	1B full; 200M / 320M distilled	Yes (export)	Distilled + INT8 [VALIDATE]
IndicConformer ASR	Yes	22 incl. Santali ; Hindi	MIT	120M/lang encoder; 600M multi	Yes	Yes, INT8 [VALIDATE]
Indic Parler-TTS	Yes	Santali official, named voices; ~298 h Santali in corpus	Apache-2.0 (verify)	Parler-Mini class (~0.9B)	Cloud only	No
sherpa-onnx	Yes	Runtime, language-agnostic	Apache-2.0	Library	Yes	Yes
ONNX Runtime Mobile	Yes	—	MIT	Library	Yes	Yes

Resource	Open	Languages relevant here	Licence	Size	Offline	2 GB feasible
CTranslate2	Yes	—	MIT	Library	Yes	[VALIDATE] on Android
Bhashini / ULCA	Platform	Multilingual conformer ASR service lists Santali	Gov platform ToS	Cloud	No	n/a
Adi Vaani (MoTA)	App	Santali, Bhili, Mundari , Gondi	Gov	App	Offline mode reported	n/a
AdiBhashaa	Yes	Bhili, Mundari , Gondi, Santali	[VALIDATE]	Corpus	n/a	n/a
Silerio VAD	Yes	—	MIT	~2 M	Yes	Yes
Noto Sans Ol Chiki	Yes	Santali script	SIL OFL	~100 KB	Yes	Yes
NLLB-200	Yes	Broad	CC-BY-NC-4.0	600M–3.3B	Yes	600M distilled marginal

42. Research References

Government / policy

- NIPUN Bharat Guidelines, Dept. of School Education & Literacy, MoE — <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2021/jul/doc20217531.pdf>
- PIB, NIPUN Bharat launch (Lakshya Soochi, Balvatika–Grade 3) — <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1732830>
- PIB, Ministry of Tribal Affairs launches Adi Vaani beta — <https://www.pib.gov.in/PressReleasePage.aspx?PRID=2162846>
- PTI/Careers360, Jharkhand minister on PALASH scope and districts (Jan 2026) — <https://news.careers360.com/need-include-tribal-regional-languages-in-primary-education-jharkhand-minister>
- MLE India, PALASH programme overview — <http://www.mle-india.net/2024/10/palash-multilingual-education-program.html>
- CALA 2019 (SOAS), M-TALL / Bhasha Puliya / MTB-MLE textbook history in Jharkhand — <https://glocal.soas.ac.uk/cala2019-388/>

Models and toolkits

- Gala et al., *IndicTrans2: Towards High-Quality and Accessible MT for all 22 Scheduled Indian Languages*, TMLR 2023 — <https://openreview.net/pdf?id=vfT4YuzAYA> · repo <https://github.com/AI4Bharat/IndicTrans2>
- IndicTrans2 distilled 200M — <https://huggingface.co/ai4bharat/indictrans2-en-indic-dist-200M> · indic-indic 320M — <https://huggingface.co/ai4bharat/indictrans2-indic-indic-dist-320M>
- IndicConformer ASR (incl. Santali) — <https://github.com/AI4Bharat/IndicConformerASR> · 600M multilingual — <https://huggingface.co/ai4bharat/indic-conformer-600m-multilingual>
- Indic Parler-TTS — <https://huggingface.co/ai4bharat/indic-parler-tts> · pretrained (per-language hours) — <https://huggingface.co/ai4bharat/indic-parler-tts-pretrained>
- sherpa-onnx — <https://github.com/k2-fsa/sherpa-onnx> · Flutter package — https://pub.dev/packages/sherpa_onnx · model zoo — https://k2-fsa.github.io/sherpa/onnx/pretrained_models/offline-transducer/index.html
- CTranslate2 — <https://github.com/OpenNMT/CTranslate2>
- On-device IndicTrans2 INT8 ONNX, measured latency and memory (third-party) — <https://huggingface.co/mobilebytesensei/betterflow-indictrans2-indic-en-200m-int8>

Datasets and benchmarks

- Singh & Kumar, *AdiBhashaa: A Community-Curated Benchmark for MT into Indian Tribal Languages*, arXiv:2512.04765 — <https://arxiv.org/abs/2512.04765> · dataset <https://huggingface.co/datasets/misniitdelhi/AdiBhasha>
- *Evaluating IndicTrans2 and ByT5 for English–Santali MT using the OI Chiki Script*, MMLoSo 2025 — <https://aclanthology.org/2025.mmloso-1.9.pdf>
- Bhashini / ULCA — <https://github.com/bhashini-dibd/ulca> · API docs <https://bhashini.gitbook.io/bhashini-apis> · model/service list <https://dibd-bhashini.gitbook.io/bhashini-apis/available-models-for-usage>

Language background

- Santali language and OI Chiki script (Unicode U+1C50–U+1C7F; official in Jharkhand) — https://en.wikipedia.org/wiki/Santali_language · https://en.wikipedia.org/wiki/OI_Chiki_script

43. Future Extensions

Teacher-facing pronunciation coaching (the tool teaches *the teacher* the language over time — arguably the highest-impact long-term feature, since it works toward its own obsolescence) · adaptive lesson sequencing from in-app signals · offline speech-based formative assessment of child reading fluency against FLN targets · community story collection (children's families contribute local stories, reviewed, added to the corpus) · OI Chiki handwriting practice with on-device stroke recognition · a Jharkhand tribal-language education corpus published openly with community credit · extension to the four PALASH regional languages (Khortha, Nagpuri, Panchpargania, Kurmali) · cross-state reuse for Odisha, Chhattisgarh, Madhya Pradesh, West Bengal, where the same languages and the same teacher-shortage pattern exist.

Recommendations

1. Build Santali. Only Santali. It is the single choice supported by published open models across MT, ASR and TTS, and that is a consequence of its Scheduled Language status, not of anyone's preference. A Ho or Mundari prototype would be a phrase bank with a translation UI bolted on. State the reason in the pitch — the language-coverage evidence table is itself a strong argument that the team did real research.

2. Make the verified-first architecture the headline, not the AI. Every competing team will demo a translation. The differentiator that survives a hard question from a judge is: *no AI-generated tribal-language content reaches a child without a native speaker approving it, and the teacher can always see which is which*. Build the verification chip on day one and never ship a screen without it.

3. Treat the phrase bank as the product's spine. It delivers the 3-second requirement, it works when every model fails, it is what native speakers can actually review at scale, and it grows with use. Start the elicitation session with real teachers in Week 1 — it has the longest lead time of anything in the plan and no amount of engineering can compensate for starting it in Week 6.

4. Buy the 2 GB tablet in Week 1 and publish your numbers, including the bad ones. The §14.4 datapoint says a naive on-device MT call takes ~5.9 s on hardware better than the target. Either the team beats that with greedy decoding and a better runtime, or it scopes the 3-second guarantee to the fast path and says so. Both are respectable. Discovering it during the demo is not.

5. Do not chase on-device TTS for the MVP. Cached, human-QA'd audio is better pedagogy (children imitate pronunciation), better latency, and lower risk. Frame synthesis as the scaling mechanism and native recording as the destination.

6. Get the official NIPUN codes and the Jharkhand-adapted Lakshya Soochi before the finals if at all possible. Placeholders are honest, but real codes queried live from the database is the difference between "aligned to NIPUN Bharat" as a slide and as a demonstrable property. The state adaptation, not the national list, is what matters.

7. Approach Adi Vaani's consortium and JEPC as collaborators, not competitors. The consortium already works with Jharkhand TRIs and covers Mundari; JEPC owns validated PALASH content and the reviewer network. A prototype that plugs into existing government

infrastructure is far more deployable than one that duplicates it — and "we build the classroom layer on top of Bhashini and Adi Vaani" is a much stronger claim than "we built our own translator."

8. Keep the licence manifest current from Week 1. NLLB's non-commercial licence is exactly the kind of detail that kills a government deployment after the demo is won. Verify every model card yourself; never inherit a licence claim from a blog post or from this document.

End of PRD v1.0. Every figure herein is a target or a cited fact; none is a measurement of this system, because this system does not exist yet. Replace the targets with measurements as they land, and keep the distinction visible.